

Abstract I

template_autopoiesis is a combinatoric grammar that **deterministically generates whole runnable projects**. Given a single integer seed and a grammar of orthogonal slots, the system produces a fully-materialized child project — complete with kernel source, tests, analysis entry-point, and a manuscript stub — whose every byte is traceable to that seed.

Generation pipeline

flowchart LR

```
G[Grammar\nconfig.yaml] --> E[Expand\nseed → Spec]  
E --> M[Materialize\nSpec → Child]  
M --> V[Verify\ntree-hash check]  
V --> S[Seal\nQR provenance]
```

Abstract II

Grammar product space

flowchart TB

A[primitive_domain: 5] --> EP[Effective product: 45 cells]

T[track: 3] --> EP

S[section_set: 3] --> EP

EP --> TP[Total product: 360 cells]

R[3 reserved slots, 2 options each] --> TP

- ▶ **Domain count:** 5
- ▶ **Effective product size:** 45
- ▶ **Total product size:** 360
- ▶ **Reserved slots:** 3 (figure_profile, qr_profile, integrity_profile)
- ▶ **Grammar hash:** f84a8f9dbcb18e37
- ▶ **Tests:** 493 · **Coverage:** 96.28%